Kernel: Python 3 (CoCalc)

TITLE: Analyzing City Service Response Efficiency in Washington, D.C. (2025)

SOURCE: 311 City Service Requests

DESCRIPTION: The dataset used in this project is Washington, D.C.'s 2025 311 City Service Requests dataset, which contains records of non-emergency service requests submitted by residents. It includes information such as service type, location, priority, and request dates. The dataset originally had 440,600 rows and 33 columns, and after cleaning, 19 key columns were kept for analysis. This dataset is useful for understanding city service demand and response patterns across different areas of Washington, D.C.

In [3]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

In [4]:
```
df = pd.read_csv("service_requests.csv", low_memory=False)
df
```

Out[4]:

	X	Y	SERVICEREQUESTID	STREETADDRESS	CITY	STATE
0	401366.06	134524.87	25-00000002	1438 K STREET SE	WASHINGTON	DC
1	396942.87	140394.80	25-00000003	1511 PARK ROAD NW	WASHINGTON	DC
2	393757.08	140604.28	25-00000004	3614 MACOMB STREET NW	WASHINGTON	DC
3	401561.80	133400.44	25-00000005	1728 16TH STREET SE	WASHINGTON	DC
4	399909.21	137153.67	25-00000007	327 K STREET NE	WASHINGTON	DC
...	...	...	...	...	...	...
440595	405313.27	136605.71	25-00790590	4409 FOOTE STREET NE	WASHINGTON	DC

	X	Y	SERVICEREQUESTID	STREETADDRESS	CITY	STATE
440596	397986.86	138741.56	25-00790594	1924 8TH STREET NW	WASHINGTON	DC
440597	405900.42	137525.60	26-00002433	1126 49TH STREET NE	WASHINGTON	DC
440598	400922.08	135939.61	26-00002483	136 TENNESSEE AVENUE NE	WASHINGTON	DC
440599	401467.44	137402.93	26-00028502	1356 LEVIS STREET NE	WASHINGTON	DC

440600 rows × 33 columns

In [5]:
```
df.shape
```

Out[5]: (440600, 33)

In [6]:
```
df.head(5)
```

Out[6]:

	X	Y	SERVICEREQUESTID	STREETADDRESS	CITY	STATE	ZIPCC
0	401366.06	134524.87	25-00000002	1438 K STREET SE	WASHINGTON	DC	20003
1	396942.87	140394.80	25-00000003	1511 PARK ROAD NW	WASHINGTON	DC	20010
2	393757.08	140604.28	25-00000004	3614 MACOMB STREET NW	WASHINGTON	DC	20016
3	401561.80	133400.44	25-00000005	1728 16TH STREET SE	WASHINGTON	DC	20020

	X	Y	SERVICEREQUESTID	STREETADDRESS	CITY	STATE	ZIPCC
4	399909.21	137153.67	25-00000007	327 K STREET NE	WASHINGTON	DC	20002

5 rows × 33 columns

In [7]:
```
df.columns
```

Out[7]:
```
Index(['X', 'Y', 'SERVICEREQUESTID', 'STREETADDRESS', 'CITY', 'STATE',
       'ZIPCODE', 'SERVICECODE', 'SERVICECODEDESCRIPTION',
       'SERVICETYPECODEDESCRIPTION', 'ORGANIZATIONACRONYM', 'SERVICECALLCOUNT',
       'ADDDATE', 'RESOLUTIONDATE', 'SERVICEDUEDATE', 'SERVICEORDERDATE',
       'INSPECTIONFLAG', 'INSPECTIONDATE', 'INSPECTORNAME',
       'SERVICEORDERSTATUS', 'STATUS_CODE', 'PRIORITY', 'DETAILS', 'WARD',
       'XCOORD', 'YCOORD', 'LATITUDE', 'LONGITUDE', 'MARADDRESSREPOSITORYID',
       'GLOBALID', 'CREATED', 'EDITED', 'OBJECTID'],
      dtype='object')
```

In [8]:
```
df.info()
```

Out[8]:
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 440600 entries, 0 to 440599
Data columns (total 33 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   X                           440600 non-null  float64
 1   Y                           440600 non-null  float64
 2   SERVICEREQUESTID            440600 non-null  object
 3   STREETADDRESS               437937 non-null  object
 4   CITY                        440600 non-null  object
 5   STATE                       440600 non-null  object
 6   ZIPCODE                     440501 non-null  float64
 7   SERVICECODE                 440600 non-null  object
 8   SERVICECODEDESCRIPTION      440600 non-null  object
 9   SERVICETYPECODEDESCRIPTION  417735 non-null  object
 10  ORGANIZATIONACRONYM         440445 non-null  object
 11  SERVICECALLCOUNT            440600 non-null  int64
 12  ADDDATE                     440600 non-null  object
 13  RESOLUTIONDATE              434480 non-null  object
 14  SERVICEDUEDATE              440586 non-null  object
 15  SERVICEORDERDATE            440600 non-null  object
 16  INSPECTIONFLAG              0 non-null       float64
 17  INSPECTIONDATE              0 non-null       float64
 18  INSPECTORNAME               0 non-null       float64
 19  SERVICEORDERSTATUS          440600 non-null  object
 20  STATUS_CODE                 0 non-null       float64
 21  PRIORITY                    440580 non-null  object
 22  DETAILS                     775 non-null     object
 23  WARD                        440590 non-null  object
 24  XCOORD                      440600 non-null  float64
 25  YCOORD                      440600 non-null  float64
 26  LATITUDE                    440600 non-null  float64
```

```
 27  LONGITUDE                   440600 non-null  float64
 28  MARADDRESSREPOSITORYID      440600 non-null  int64
 29  GLOBALID                    440600 non-null  object
 30  CREATED                     0 non-null       float64
 31  EDITED                      0 non-null       float64
 32  OBJECTID                    440600 non-null  int64
dtypes: float64(13), int64(3), object(17)
memory usage: 110.9+ MB
```

In [9]:

```
print(df.columns.tolist())
```

Out[9]:

```
['X', 'Y', 'SERVICEREQUESTID', 'STREETADDRESS', 'CITY', 'STATE', 'ZIPCODE',
'SERVICECODE', 'SERVICECODEDESCRIPTION', 'SERVICETYPECODEDESCRIPTION',
'ORGANIZATIONACRONYM', 'SERVICECALLCOUNT', 'ADDDATE', 'RESOLUTIONDATE',
'SERVICEDUEDATE', 'SERVICEORDERDATE', 'INSPECTIONFLAG', 'INSPECTIONDATE',
'INSPECTORNAME', 'SERVICEORDERSTATUS', 'STATUS_CODE', 'PRIORITY', 'DETAILS',
'WARD', 'XCOORD', 'YCOORD', 'LATITUDE', 'LONGITUDE', 'MARADDRESSREPOSITORYID',
'GLOBALID', 'CREATED', 'EDITED', 'OBJECTID']
```

In [10]:

```
cols_to_drop = [
    "INSPECTIONFLAG",
    "INSPECTIONDATE",
    "INSPECTORNAME",
    "STATUS_CODE",
    "CREATED",
    "EDITED"
]

df = df.drop(columns=[col for col in cols_to_drop if col in df.columns])
```

In [11]:

```
df
```

Out[11]:

	X	Y	SERVICEREQUESTID	STREETADDRESS	CITY	STATE
0	401366.06	134524.87	25-00000002	1438 K STREET SE	WASHINGTON	DC
1	396942.87	140394.80	25-00000003	1511 PARK ROAD NW	WASHINGTON	DC
2	393757.08	140604.28	25-00000004	3614 MACOMB STREET NW	WASHINGTON	DC
3	401561.80	133400.44	25-00000005	1728 16TH STREET SE	WASHINGTON	DC

	X	Y	SERVICEREQUESTID	STREETADDRESS	CITY	STATE
4	399909.21	137153.67	25-00000007	327 K STREET NE	WASHINGTON	DC
...	...	...	...	...	...	...
440595	405313.27	136605.71	25-00790590	4409 FOOTE STREET NE	WASHINGTON	DC
440596	397986.86	138741.56	25-00790594	1924 8TH STREET NW	WASHINGTON	DC
440597	405900.42	137525.60	26-00002433	1126 49TH STREET NE	WASHINGTON	DC
440598	400922.08	135939.61	26-00002483	136 TENNESSEE AVENUE NE	WASHINGTON	DC
440599	401467.44	137402.93	26-00028502	1356 LEVIS STREET NE	WASHINGTON	DC

440600 rows × 27 columns

In [12]:
```
print(df.isnull().sum())
```

Out[12]:
```
X                                  0
Y                                  0
SERVICEREQUESTID                   0
STREETADDRESS                   2663
CITY                               0
STATE                              0
ZIPCODE                           99
SERVICECODE                        0
SERVICECODEDESCRIPTION             0
SERVICETYPECODEDESCRIPTION     22865
ORGANIZATIONACRONYM              155
SERVICECALLCOUNT                   0
ADDDATE                            0
RESOLUTIONDATE                  6120
SERVICEDUEDATE                    14
SERVICEORDERDATE                   0
SERVICEORDERSTATUS                 0
PRIORITY                          20
```

```
DETAILS                        439825
WARD                               10
XCOORD                              0
YCOORD                              0
LATITUDE                            0
LONGITUDE                           0
MARADDRESSREPOSITORYID              0
GLOBALID                            0
OBJECTID                            0
dtype: int64
```

In [13]:

```
drop_cols = [
    "X","Y","XCOORD","YCOORD",
    "GLOBALID","OBJECTID",
    "MARADDRESSREPOSITORYID",
    "SERVICEREQUESTID",
    "SERVICECODE",
    "DETAILS"
]

df = df.drop(columns=drop_cols, errors="ignore")
```

In [14]:

```
df
```

Out[14]:

	STREETADDRESS	CITY	STATE	ZIPCODE	SERVICECODEDESCRIPTION	SE
0	1438 K STREET SE	WASHINGTON	DC	20003.0	Bulk Collection	SV Ac
1	1511 PARK ROAD NW	WASHINGTON	DC	20010.0	Parking Enforcement	PE Ma
2	3614 MACOMB STREET NW	WASHINGTON	DC	20016.0	Bulk Collection	SV Ac
3	1728 16TH STREET SE	WASHINGTON	DC	20020.0	Bulk Collection	SV Ac
4	327 K STREET NE	WASHINGTON	DC	20002.0	Bulk Collection	SV Ac
...	...	...	...	...	...	...
440595	4409 FOOTE STREET NE	WASHINGTON	DC	20019.0	Bulk Collection	SV Ac
440596	1924 8TH STREET NW	WASHINGTON	DC	20001.0	Parking Enforcement	PE Ma
440597	1126 49TH STREET NE	WASHINGTON	DC	20019.0	Trash Cart - Delivery	SV Ac
440598	136 TENNESSEE AVENUE NE	WASHINGTON	DC	20002.0	Recycling- Information Request	SV Ac
440599	1356 LEVIS STREET NE	WASHINGTON	DC	20002.0	Trash Cart - Delivery	SV Ac

440600 rows × 17 columns

In [15]:
```
df["STREETADDRESS"] = df["STREETADDRESS"].fillna("Unknown")
df["ZIPCODE"] = df["ZIPCODE"].fillna(df["ZIPCODE"].mode()[0])
df["SERVICETYPECODEDESCRIPTION"] =
df["SERVICETYPECODEDESCRIPTION"].fillna("Unknown")
df["ORGANIZATIONACRONYM"] = df["ORGANIZATIONACRONYM"].fillna("Unknown")
df["PRIORITY"] = df["PRIORITY"].fillna("Unknown")
df["WARD"] = df["WARD"].fillna("Unknown")
```

In [16]:
```
df
```

Out[16]:

	STREETADDRESS	CITY	STATE	ZIPCODE	SERVICECODEDESCRIPTION	SE
0	1438 K STREET SE	WASHINGTON	DC	20003.0	Bulk Collection	SV Ac
1	1511 PARK ROAD NW	WASHINGTON	DC	20010.0	Parking Enforcement	PE Ma
2	3614 MACOMB STREET NW	WASHINGTON	DC	20016.0	Bulk Collection	SV Ac
3	1728 16TH STREET SE	WASHINGTON	DC	20020.0	Bulk Collection	SV Ac
4	327 K STREET NE	WASHINGTON	DC	20002.0	Bulk Collection	SV Ac
...	...	...	...	...	...	...
440595	4409 FOOTE STREET NE	WASHINGTON	DC	20019.0	Bulk Collection	SV Ac
440596	1924 8TH STREET NW	WASHINGTON	DC	20001.0	Parking Enforcement	PE Ma
440597	1126 49TH STREET NE	WASHINGTON	DC	20019.0	Trash Cart - Delivery	SV Ac
440598	136 TENNESSEE AVENUE NE	WASHINGTON	DC	20002.0	Recycling- Information Request	SV Ac
440599	1356 LEVIS STREET NE	WASHINGTON	DC	20002.0	Trash Cart - Delivery	SV Ac

440600 rows × 17 columns

In [17]:
```
print(df.isnull().sum())
```

Out[17]:
```
STREETADDRESS                    0
CITY                             0
STATE                            0
ZIPCODE                          0
SERVICECODEDESCRIPTION           0
SERVICETYPECODEDESCRIPTION       0
ORGANIZATIONACRONYM              0
SERVICECALLCOUNT                 0
ADDDATE                          0
RESOLUTIONDATE                6120
```

```
SERVICEDUEDATE                 14
SERVICEORDERDATE                0
SERVICEORDERSTATUS              0
PRIORITY                        0
WARD                            0
LATITUDE                        0
LONGITUDE                       0
dtype: int64
```

In [18]:

```
df["Resolved"] = df["RESOLUTIONDATE"].notna()
df
```

Out[18]:

	STREETADDRESS	CITY	STATE	ZIPCODE	SERVICECODEDESCRIPTION	SE
0	1438 K STREET SE	WASHINGTON	DC	20003.0	Bulk Collection	SV Ac
1	1511 PARK ROAD NW	WASHINGTON	DC	20010.0	Parking Enforcement	PE Ma
2	3614 MACOMB STREET NW	WASHINGTON	DC	20016.0	Bulk Collection	SV Ac
3	1728 16TH STREET SE	WASHINGTON	DC	20020.0	Bulk Collection	SV Ac
4	327 K STREET NE	WASHINGTON	DC	20002.0	Bulk Collection	SV Ac
...	...	...	...	...	...	...
440595	4409 FOOTE STREET NE	WASHINGTON	DC	20019.0	Bulk Collection	SV Ac
440596	1924 8TH STREET NW	WASHINGTON	DC	20001.0	Parking Enforcement	PE Ma
440597	1126 49TH STREET NE	WASHINGTON	DC	20019.0	Trash Cart - Delivery	SV Ac
440598	136 TENNESSEE AVENUE NE	WASHINGTON	DC	20002.0	Recycling- Information Request	SV Ac
440599	1356 LEVIS STREET NE	WASHINGTON	DC	20002.0	Trash Cart - Delivery	SV Ac

440600 rows × 18 columns

In [19]:

```
df["RESOLUTIONDATE"].isna().sum()
```

Out[19]:

```
np.int64(6120)
```

In [20]:

```
df["Resolved"].value_counts() #To show how many cases were resolved (true) and
unresolved (False)
```

Out[20]:

```
Resolved
True     434480
False      6120
Name: count, dtype: int64
```

In [21]:
```
df.groupby("WARD")["Resolved"].mean() #This shows a percentage of the cases that were resolved by ward. For example, Ward 1 had about 98% cases resolved.
```

Out[21]:
```
WARD
Unknown    1.000000
Ward 1     0.986040
Ward 2     0.980948
Ward 3     0.981179
Ward 4     0.988323
Ward 5     0.984780
Ward 6     0.987786
Ward 7     0.986835
Ward 8     0.992039
Name: Resolved, dtype: float64
```

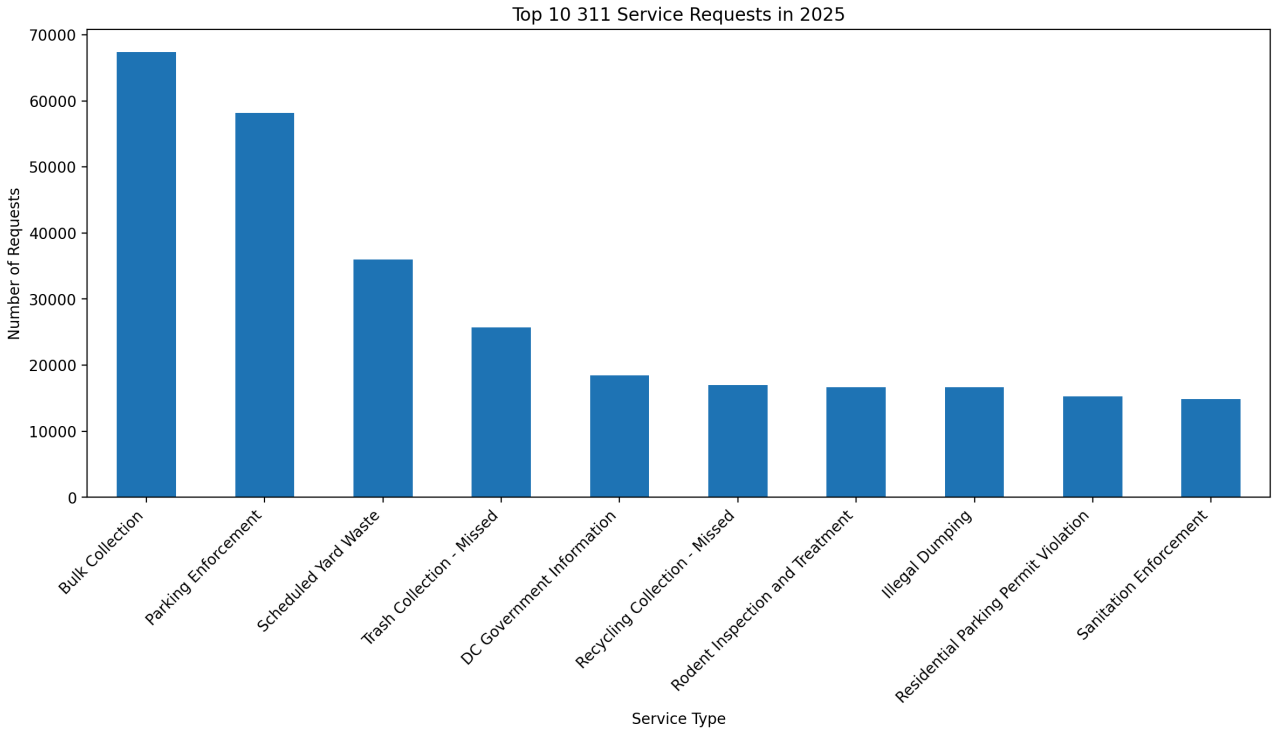
What are the most common types of service requests?

In [22]:
```
top_requests = df["SERVICECODEDESCRIPTION"].value_counts().head(10)
top_requests
```

Out[22]:
```
SERVICECODEDESCRIPTION
Bulk Collection                          67414
Parking Enforcement                      58149
Scheduled Yard Waste                     36011
Trash Collection - Missed                25722
DC Government Information                18455
Recycling Collection - Missed            16985
Rodent Inspection and Treatment          16684
Illegal Dumping                          16678
Residential Parking Permit Violation     15283
Sanitation Enforcement                   14853
Name: count, dtype: int64
```

In [23]:
```
top_requests.plot(kind="bar")

plt.title("Top 10 311 Service Requests in 2025")
plt.xlabel("Service Type")
plt.ylabel("Number of Requests")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()

plt.show()
```

Out[23]:



The distribution of 311 service requests was analyzed to identify the most frequently reported issues in 2025. A bar chart of the top 10 service types shows that Bulk Collection is the most common concern reported by residents. This suggests that this issue represents the primary areas of public service demand.

Where are the problems happening? (WARD analysis)

In [24]:

```
ward_rate = df["WARD"].value_counts(normalize=True) * 100
ward_rate.sort_index()
```

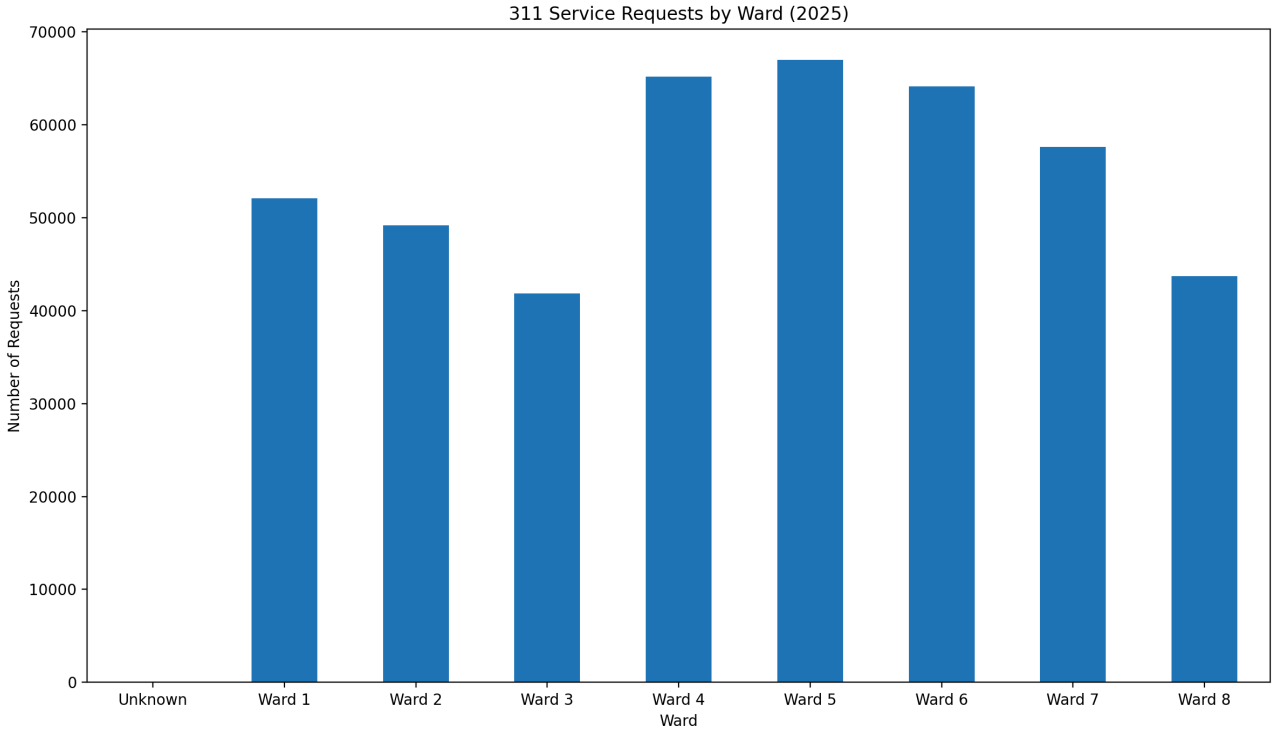
Out[24]:

```
WARD
Unknown     0.002270
Ward 1     11.819337
Ward 2     11.162279
Ward 3      9.490695
Ward 4     14.791194
Ward 5     15.195188
Ward 6     14.550159
Ward 7     13.067635
Ward 8      9.921244
Name: proportion, dtype: float64
```

In [25]:

```
ward_counts = df["WARD"].value_counts().sort_index()

ward_counts.plot(kind="bar")

plt.title("311 Service Requests by Ward (2025)")
plt.xlabel("Ward")
plt.ylabel("Number of Requests")
plt.xticks(rotation=0)

plt.tight_layout()
plt.show()
```

Out[25]:



This shows Ward 5 generates the most 311 service requests

How long does it take to resolve service requests?

In [26]:

```
df["ADDDATE"] = pd.to_datetime(df["ADDDATE"], errors="coerce")
df["RESOLUTIONDATE"] = pd.to_datetime(df["RESOLUTIONDATE"], errors="coerce") #
To convert date columns in text strings to actual dates that can be used for
calculations.
```

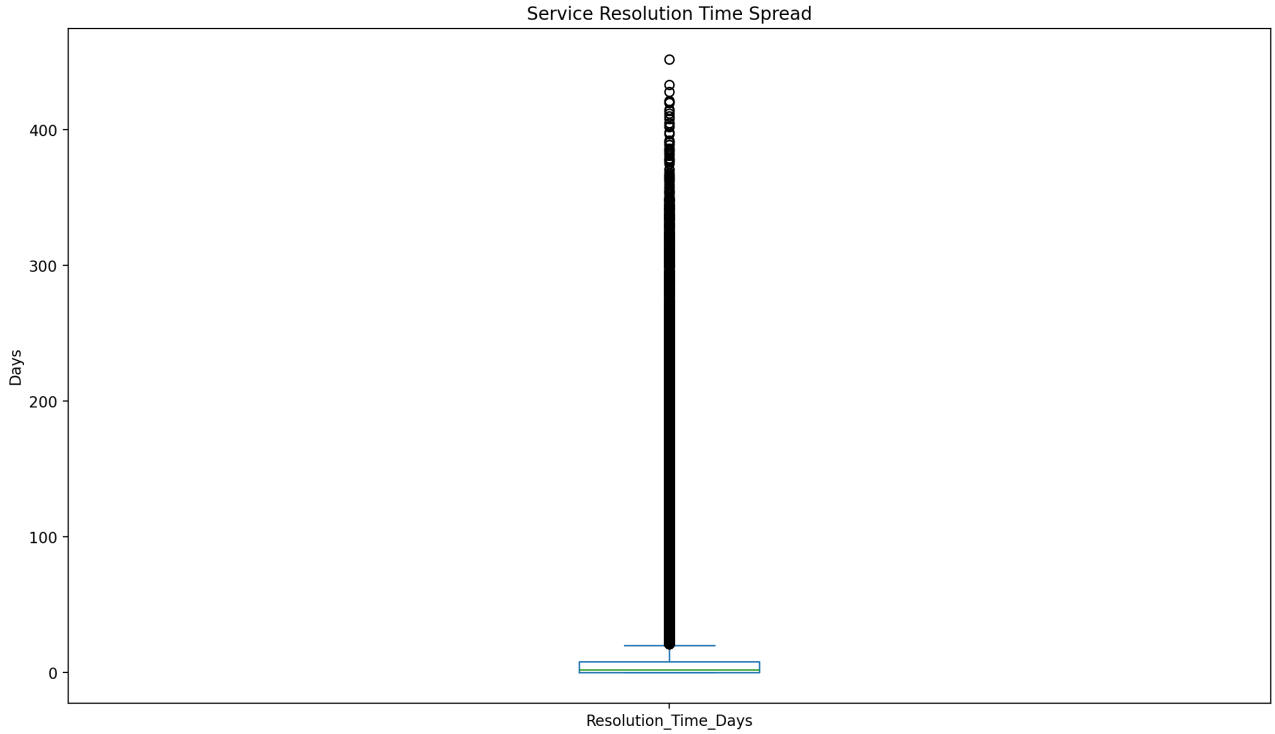
In [27]:

```
df["Resolution_Time_Days"] = (
    df["RESOLUTIONDATE"] - df["ADDDATE"]
).dt.days
```

In [28]:

```
df["Resolution_Time_Days"].dropna().plot(kind="box")

plt.title("Service Resolution Time Spread")
plt.ylabel("Days")

plt.tight_layout()
plt.show()
```

Out[28]:



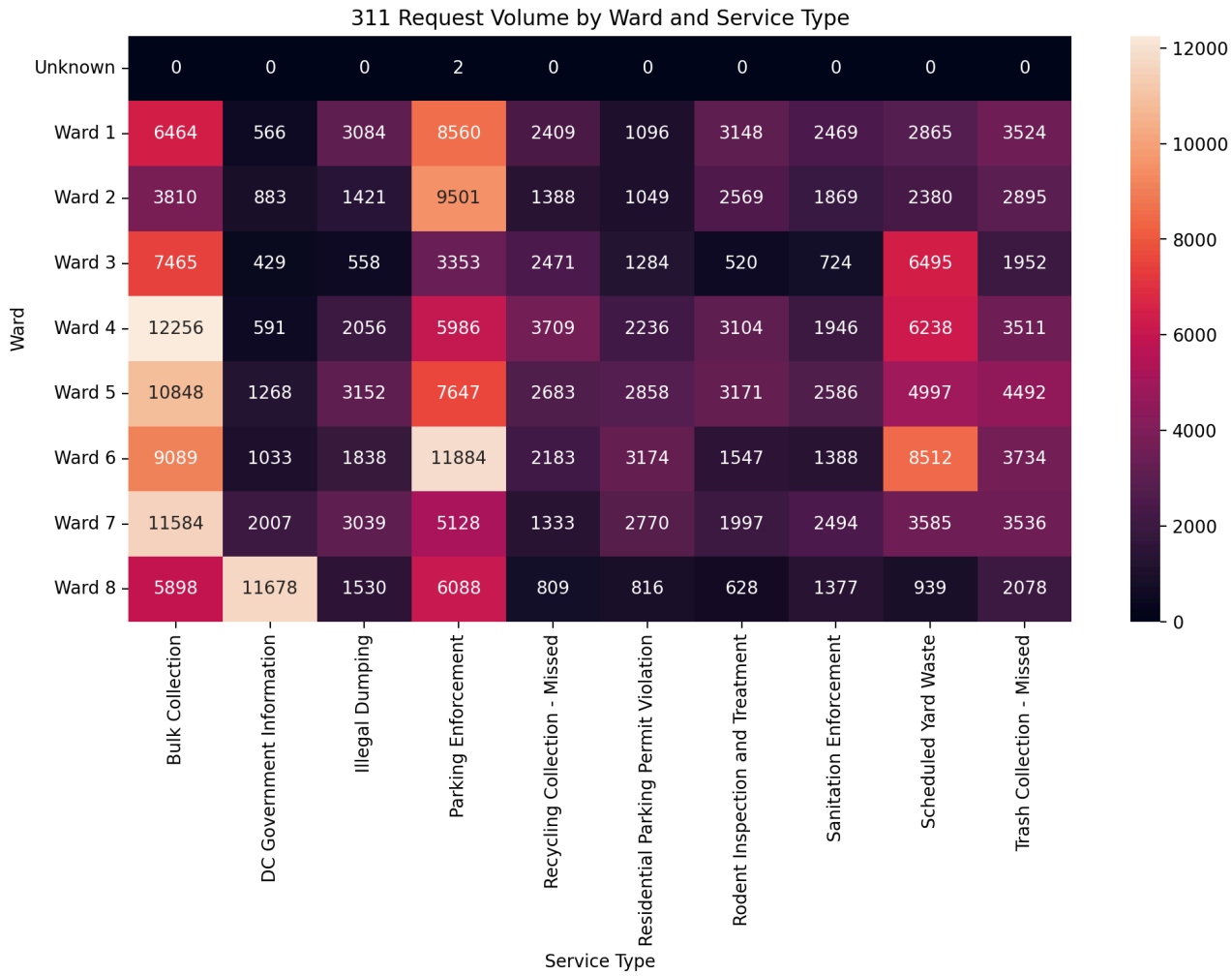
The plot shows majority of the cases were resolved in a very quick manner. The top circles showed some cases took some more time to get resolved.

Which wards report which problems the most?

In [29]:

```
top_services = df["SERVICECODEDESCRIPTION"].value_counts().head(10).index

# filter to only top services
heat_df = df[df["SERVICECODEDESCRIPTION"].isin(top_services)]

# create grouped table
heat_data = heat_df.groupby(
    ["WARD", "SERVICECODEDESCRIPTION"]
).size().unstack(fill_value=0)

# plot heatmap
plt.figure(figsize=(12, 6))
sns.heatmap(heat_data, annot=True, fmt="d")

plt.title("311 Request Volume by Ward and Service Type")
plt.xlabel("Service Type")
plt.ylabel("Ward")
plt.show()
```

Out[29]:

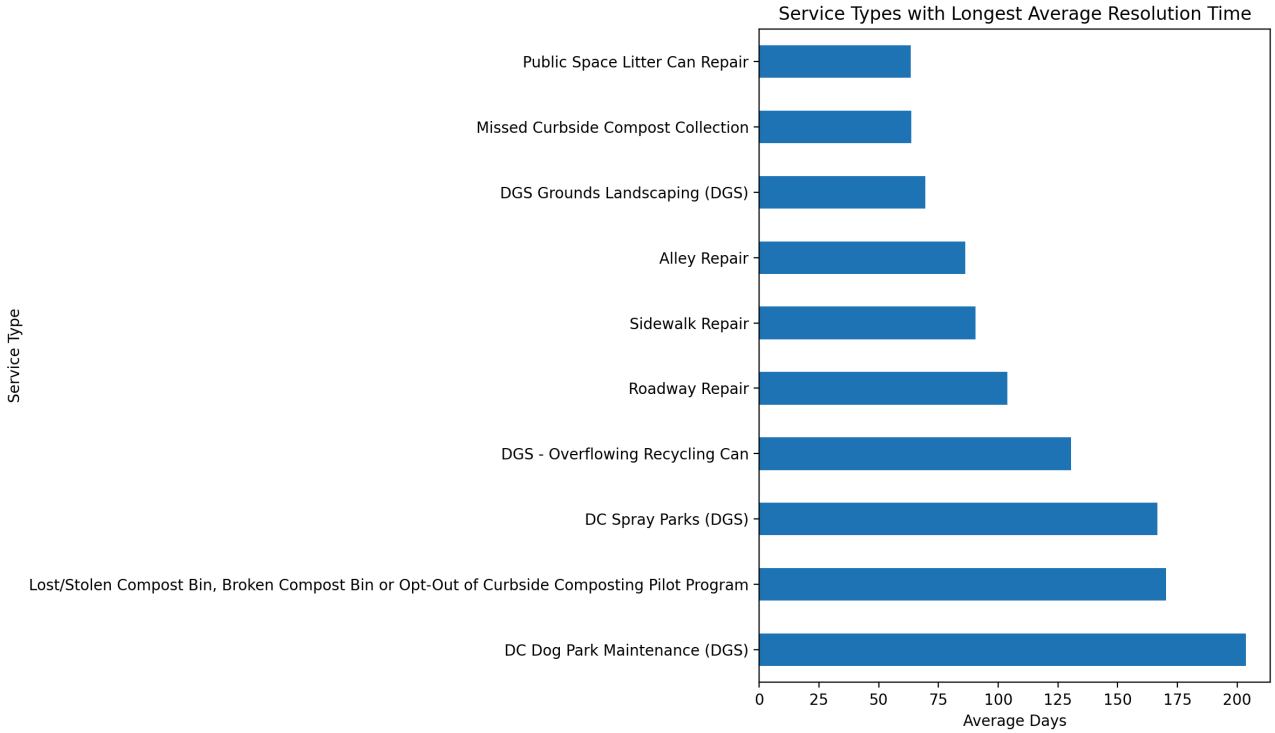


Which service types take the longest to resolve?

In [30]:

```
filtered = df.groupby("SERVICECODEDESCRIPTION").filter(lambda x: len(x) >= 100)

avg_resolution = (
    filtered.groupby("SERVICECODEDESCRIPTION")["Resolution_Time_Days"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

avg_resolution.plot(kind="barh")

plt.title("Service Types with Longest Average Resolution Time")
plt.xlabel("Average Days")
plt.ylabel("Service Type")

plt.tight_layout()
plt.show()
```

Out[30]:



In [31]:

```
df
```

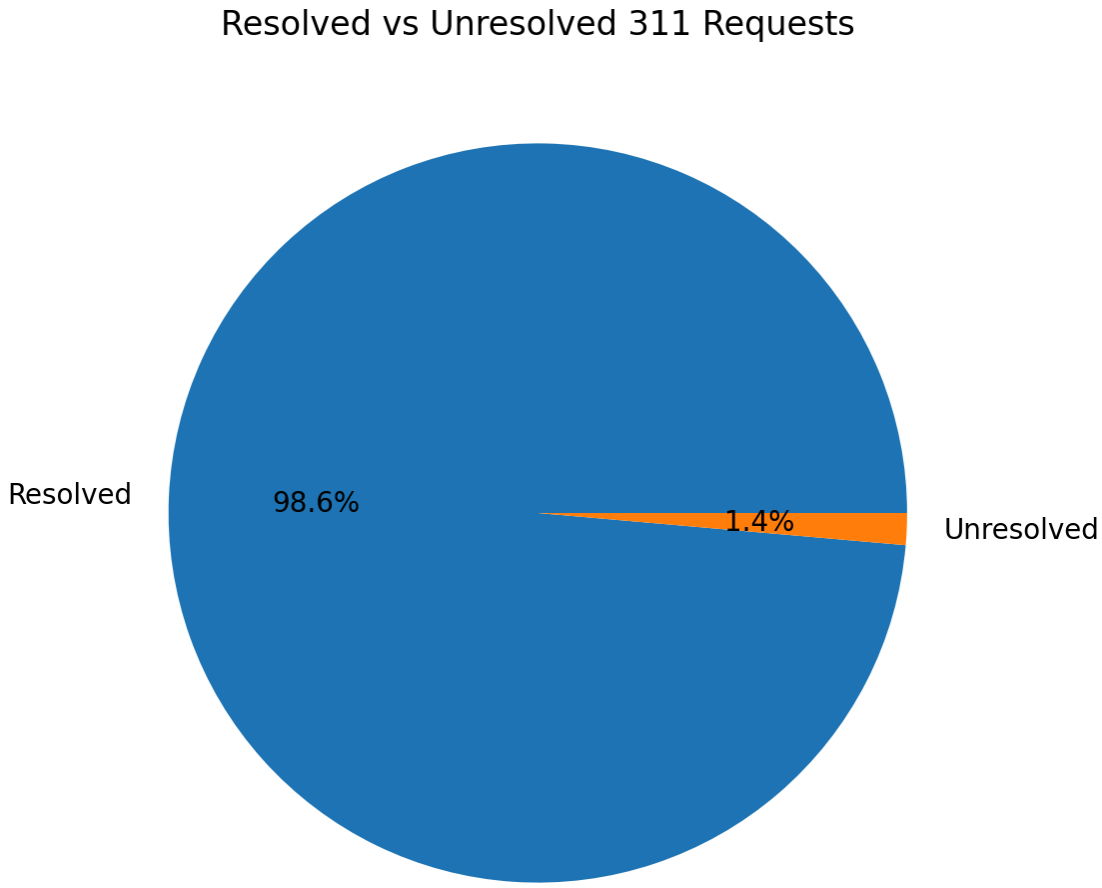
Out[31]:

	STREETADDRESS	CITY	STATE	ZIPCODE	SERVICECODEDESCRIPTION	SE
0	1438 K STREET SE	WASHINGTON	DC	20003.0	Bulk Collection	SV Ac
1	1511 PARK ROAD NW	WASHINGTON	DC	20010.0	Parking Enforcement	PE Ma
2	3614 MACOMB STREET NW	WASHINGTON	DC	20016.0	Bulk Collection	SV Ac
3	1728 16TH STREET SE	WASHINGTON	DC	20020.0	Bulk Collection	SV Ac
4	327 K STREET NE	WASHINGTON	DC	20002.0	Bulk Collection	SV Ac
...	...	...	...	...	...	...
440595	4409 FOOTE STREET NE	WASHINGTON	DC	20019.0	Bulk Collection	SV Ac
440596	1924 8TH STREET NW	WASHINGTON	DC	20001.0	Parking Enforcement	PE Ma
440597	1126 49TH STREET NE	WASHINGTON	DC	20019.0	Trash Cart - Delivery	SV Ac
440598	136 TENNESSEE AVENUE NE	WASHINGTON	DC	20002.0	Recycling- Information Request	SV Ac
440599	1356 LEVIS STREET NE	WASHINGTON	DC	20002.0	Trash Cart - Delivery	SV Ac

440600 rows × 19 columns

In [32]:
```
resolved_counts = df["Resolved"].value_counts()

plt.figure(figsize=(6,6))
plt.pie(resolved_counts, labels=["Resolved", "Unresolved"], autopct="%1.1f%%")

plt.title("Resolved vs Unresolved 311 Requests")
plt.show()
```

Out[32]:



The plot shows majority of the cases were resolved

In [33]:
```
df.duplicated().sum()
```

Out[33]: np.int64(4)

In [34]:
```
df.shape
```

Out[34]: (440600, 19)

In [35]:
```
df.to_csv('Alexander_Tewobola_final_project.csv', index=False)
```

In [ ]: